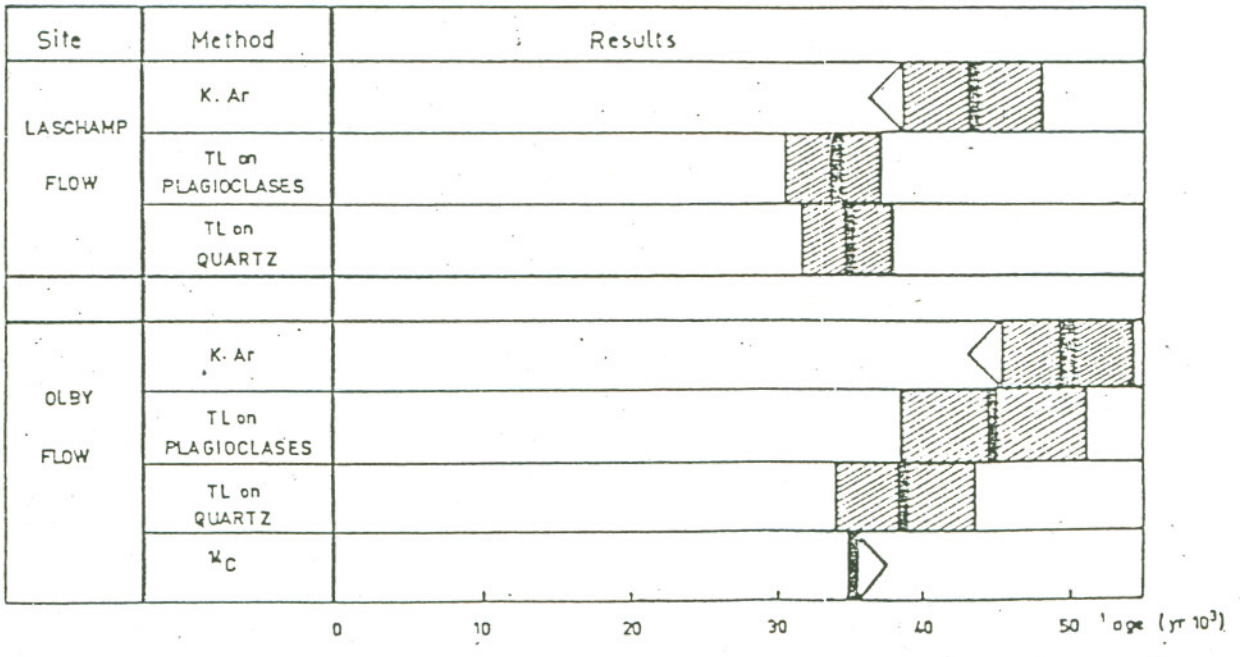
Analytical Chemistry
Cumulative Examination
February 16, 1991

The Chaine des Puys region of the Massif Central in France consists of a number of volcanoes which have been active in geologically recent times. A number of discrete lava flows can be identified with ages ranging from 8000 to more than 100,000 years. Two flows which have received particular attention are those at Laschamps and Olby which flowed at a time when the magnetic field of the earth was reversed; their ages therefore provide a unique opportunity to date the reversals. The figure below summarizes the dates obtained by Gillot and co-workers (Earth Planet Sci. Lett., 42, 444-450, 1979). Write an essay describing the principles of each method with emphasis on their strengths and weaknesses. What refinements, not used by the French, could you suggest to improve any of the data?